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MILITARY SPECIFICATION

GAGE, PRESSURE, DIAL INDICATING, MANIFOLD PRESSURE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers design requirements and all performance requirements for procurement of aneroid-type, single and dual manifold pressure gages.

1.2 Classification. The manifold pressure gages shall operate within the pressure ranges from 10 to 75 inches Hq or 15 to 100 inches Hq and shall be classified as specified in MS 28077.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification, to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-C-320 —Chromium Plating (Electrodeposited)
QQ-N-290 —Nickel Plating (Electrodeposited).
QQ-P-416 —Plating, Cadmium (Electrodeposited).
PPP-B-636 —Boxes, Fiber, Solid (For domestic shipment).

MILITARY

MIL-P-116 —Preservation, Methods of.
MIL-N-3336 —Nut, Self Locking, Instrument Mounting.
MIL-E-5272 —Environmental testing, Aeronautical and Associated Equipment, General Specification for.
MIL-C-5541 —Chemical Films for Aluminum and Aluminum Alloys.
MIL-S-6872 —Soldering Process, General Specification for.
MIL-L-6880 —Lubrication of Aircraft, General Specification for.
MIL-S-7742 —Screw Threads, Standard, Optimum Selected Series, General Specification for.
MIL-P-7936 —Parts and Equipment, Aeronautical, Preparation for Delivery.
MIL-A-8625 —Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-L-25142 —Luminescent Materiel Fluorescent.
MIL-D-70327 —Drawings, Engineering and Associated Lists.

STANDARDS

FEDERAL

FED-STD-595 —Colors.

FSC 6620

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MIL–STD–129 —Marking for Shipment and Storage.
MIL–STD–130 —Identification Marking of U.S. Military Property.
MIL–STD–143 —Specifications and Standards, Order of Precedence for the Selection of.
MIL–STD–831 —Test Reports, Preparation of.
MS28077 —Gage, Pressure, Dial Indicating, Manifold Pressure.
MS28105 —Window, Dial-Aircraft Instrument Cover, Glass.
MS33558 —Numerals and Letters, Aircraft Instrument Dial, Standard Form of.
MS33585 —Pointers, Dial, Standard Design of Aircraft Instrument.
MS33586 —Metals, Definition of Dissimilar.

(Copies of specifications and standards required by contractors in conjunction with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The gage furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualification Products List.

3.2 Selection of Government documents. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL–STD–143 except as provided in 3.2.1 and 3.2.2.

3.2.1 *Commercial parts.* Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 *Standard parts.* With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part number.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 *Critical materials.* Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 *Metals.* Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the gage may be subjected when in storage or during normal service life.

3.3.2.1 *Dissimilar metals.* Unless suitably protected against electrolytic corrosion, dissimilar metals are defined in Standard MS33586 shall not be used in intimate contact with each other.

3.3.3 *Nonmagnetic materials.* Nonmagnetic materials shall be used for all parts of the gage except where magnetic materials are essential.

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3.3.4 *Fungus-proof materials.* Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.5 *Corrosive fumes.* The materials as installed in the gage and under the service conditions specified herein shall not liberate deleterious fumes.

3.4 Design and construction. The manifold pressure gages shall conform to Standard MS28077. Gages shall be so constructed as to withstand the normal strains of jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service without failure. The operating mechanism shall consist of a sealed evacuated aneroid compartment and an indicating mechanism. Dual indicators shall have two aneroids, each connected to the respective pointer. The operating mechanism shall constitute a unit separate from the case and shall be sufficiently rugged to prevent small distortion of the case from affecting the accuracy of the gage. Unless otherwise specified, the mechanism, including the dial, shall be mounted on a metal plate which forms the back of the instrument, and which is secured to it by not less than three screws. The mechanism, including the dial, shall be removable from the rear of the case as a unit. Operation of the gage shall not be dependent upon the sealing of the case by the cover glass.

3.4.1 *Maintenance.* The design of the gage shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance using those tools and items of maintenance equipment which are normally available as commercial standards.

3.4.2 *Mechanism adjustment.* The gage shall be provided with means to adjust or correct the indication in case it becomes necessary during the service life of the instrument. The means of adjustment shall be reasonably simple but need not be external to the case. Correction adjustments shall be possible without removing the pointer or pointers. Adjustment shall not be accomplished by moving the dial.

3.4.3 *Seal.* The instrument shall be so designed that the mechanism is not exposed to fuel vapor entering from the manifold pressure connection. The mechanism shall be sealed from such exposure by a metal diaphragm, metal bellows, or similar means.

3.5 Performance. The gage shall perform satisfactorily when subjected to the tests specified in Section 4.

3.6 Case. The gage case shall conform to Standard MS38077 and shall be raintight. Reinforcement shall be provided so that the case will not be damaged when connections are tightened during gage installation.

3.6.1 *Body and mounting flange.* The body and mounting flange of the case shall be made of lightweight nonmagnetic metal, uniform in texture, and shall have a smooth surface. The body and the mounting flange shall be finished with a durable lusterless black finish Color No. 37038 of FED STD. No. 595.

3.6.2 *Cover glass.* The cover glass shall be of the size as specified in Standard MS28-105 and shall be clear and free of flaws which interfere with normal reading of the gage.

3.6.2.1 *Dial to cover glass distance.* The distance from the dial to the cover glass shall be as small as practicable and shall not exceed .125 inch for each single gage and 0.188 inch for each dual gage.

3.6.3 *Pointers.* Pointers shall conform to Standard MS33585 and be installed as listed in Standard MS28077. They shall be light and sufficiently rigid and shall be free to swing beyond either end of the range without being engaged by a mechanical stop. For the high end of the scale, means shall be provided to prevent the sector gear from becoming disengaged from the pointer pinion.

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At the end of the scale, the pointer shall be free to indicate manifold pressures down to 2 and 5 inches of mercury. Pointers shall be firmly attached to the mechanism. In the dual gage, the pointers marked with even numbers (2, 4, and 6) shall be in front of the pointers marked with odd numbers (1, 3, and 5).

3.7 **Screw threads.** Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL–S–7742.

3.8 **Interchangeability.** All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL–D–70327.

3.9 **Physical requirements.**

3.9.1 *Weight.* The weight of the gage shall not exceed 1.3 pounds for either single or dual gages.

3.9.2 *Finish.* Protective coatings and finishes which will crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.9.2.1 *Aluminum alloy parts.* Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL–A–8625. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL–C–5541.

3.9.2.2 *Iron and steel parts.* Where practicable, iron and steel parts shall be plated in accordance with Specification QQ–C–320, QQ–N–290, or QQ–P–416 as applicable.

3.9.3 *Soldering.* Soldering shall be performed in accordance with Specification MIL–S–6872.

3.9.4 *Lubrication.* Lubrication if required shall be accomplished in accordance with MIL–L–6880, except the selection of the lubricant used shall be approved by the procuring activity.

3.10 **Dial.** The dial shall be marked as shown in Figure 1. Except when fluorescent-luminescent material is required, all visible portions of the dial shall be lusterless black Color No. 37038 of FED. STD. No. 595 and all markings on the dial shall be lusterless white Color No. 37875 of FED. STD. No. 595.

3.10.1 *Fluorescent-luminescent material.* Fluorescent-luminescent material shall conform to Specification MIL–L–25142, type I or III as applicable.

3.10.2 *Marking.* All markings shall be durable to withstand usage encountered in service. The form of the letters and numerals shall conform to Standard MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.10.2.1 *Visibility of dial.* At least $\frac{1}{16}$ inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

3.11 **Identification of product.** A nameplate shall be securely attached to the gage case, and shall be marked in accordance with Standard MIL–STD–130. For Navy procurement only, the FSN number shall be omitted.

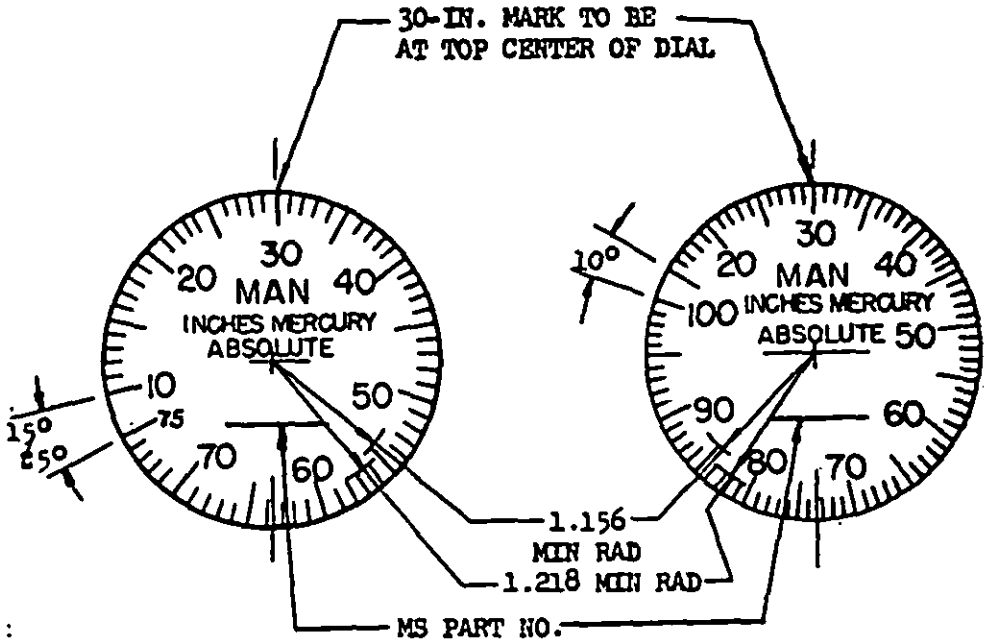
3.12 **Installation.**

3.12.1 *Mounting nuts.* The contractor shall furnish sufficient removable spring nuts conforming to Specification MIL–N–3336 to fit each mounting lug of the gage.

3.12.2 *Mounting screws.* The contractor shall furnish sufficient mounting screws for installing the gage. The screws shall be No. 6–32NC–2, round head, brass machine screws having a durable and lusterless black oxidized, or lusterless black nickel finish. The length shall be sufficient for mounting on a panel up to $\frac{3}{8}$ inch thick.

3.13 **Workmanship.** The gage, including all parts and accessories, shall be constructed and finished to produce an instrument free

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Marking	Height or length	Width of line or graduation (±.003)	Material or finish
Numerals 20, 30, 40, 50, 60 (on both gages). In addition, numerals 70, 80, 100 (on 100-inch gage).	.250	.....	Fluorescent-luminescent material or lusterless white
Lettering "Man"	.188	.....	
5- and 10-inch graduations	.219 Min	.031	
Shaded part of pointer	.....	.....	
Numerals 10 and 70 } On 75-inch gage	.250	.....	
Numeral 75 } On 75-inch gage	.125	.....	
1-inch graduations	.156 Min	.016	
Lettering "Inches Mercury Absolute"	.063	.....	Lusterless black
MS part number	.063	.....	
Background of dial	.....	.....	
Unshaded part of pointer	.....	.....	

FIGURE 1. *Dials and pointers*

from all defects which would affect proper functioning in service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the gage shall be classified as follows:

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a. Qualification tests. Qualification tests are those tests performed on gages submitted for qualifications as a satisfactory product.

b. Acceptance tests. Acceptance tests are those tests performed on gages manufactured and submitted for acceptance under contract.

4.3 **Qualification tests.** The qualification tests of the gage shall consist of all tests of this specification under paragraph 4.7 headed Test Methods, conducted in the order listed therein.

4.3.1 *Qualification test samples.* The qualification test samples shall consist of three gages of each manufacturer's part number submitted for qualification testing. After obtaining authorization for submittal from the activity responsible for qualification, the samples shall be forwarded to the testing laboratory designated in the letter of authorization. Each sample shall be plainly identified by a securely attached durable tag marked with the following information: (see 6.3).

Sample for qualification tests
Submitted by (name) (date) for qualification tests in accordance with the requirements of Specification MIL–G–5678 under authorization (reference letter authorizing test)
Gage, Single or Dual as applicable, Dial Indicating, Manifold Pressure
Manufacturer's Part No.
Name of Manufacturer

4.3.2 *Test Report.* When tests are conducted at a location other than the laboratory of the procuring activity, three copies of a test report in accordance with MIL–STD–831 shall be furnished to that activity.

4.4 **Acceptance tests.** The acceptance tests of the gages shall consist of the Individual tests, and the Sampling plans of this specification.

4.4.1 *Individual tests.* The individual tests of the gages shall consist of the following tests conducted on each gage:

Inspection
Scale error at room temperature
Friction error
Position error
Damping

4.4.2 *Sampling Plan Tests.* The sampling plan tests shall consist of Sampling Plan A and Sampling Plan B tests. Gages selected for sampling tests shall first have passed the Individual tests. Gages which have been subjected to Sampling Plan A and B tests shall not be delivered on contract until they have been refurbished and resubmitted for all the Individual tests.

4.4.2.1 *Sampling Plan A Sample Selection.* Samples shall be selected at random in accordance with the following schedule:

Quantity of gages offered for Acceptance	Quantity to be Selected for Sampling Plan A Tests
First 15	0
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

When a defective unit occurs, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs shift to one sample out of the first fifteen (when Sampling Plan B has been omitted) and proceed as indicated in paragraph 4.4.2.1.1.

4.4.2.1.1 *Sampling Plan A Tests.* Each gage selected for Sampling Plan A tests shall be subjected to the following:

Seasoning
Scale error at high temperature
Scale error-high altitude, low temperature
Overpressure
Vibration error

4.4.2.2 *Sampling Plan B Sample Selection.* Unless otherwise specified, two gages shall be selected at random from the first 15 gages produced on contract and submitted within 10 days after manufacture. These gages shall be forwarded to a laboratory designated by the procuring agency (See Section 6). Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

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Submitted by (Name) (Date) for Sample Plan B tests in accordance with (Contract No.) Manufacturer's Part Number. ———————

4.4.2.2.1 *Sampling Plan B Tests.* Each gage selected for Sampling Plan B tests shall be subjected to the following:

Sampling Plan A tests	Low temperature exposure
Vibration failure	Humidity
Magnetic effect	Life
High temperature exposure	Internal examination

4.5 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.6 Test conditions.

4.6.1 *Atmospheric Condition.* Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient 25° ±5° C
Pressure	Normal atmospheric (approx. 29.92 inches Hg)
Humidity	Room ambient up to 90% relative humidity

4.6.2 *Attitude.* Unless otherwise specified, the gage shall be tested in its level position in which the upper mounting holes are on a horizontal line and the face is vertical.

4.6.3 *Test readings.* Unless otherwise specified, before a test reading is taken, the gage shall be vibrated using an electric vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch.

4.6.4 All readings taken in subsequent tests shall use a standard mercury manometer corrected for changes in ambient (barometric) pressure at the time of testing.

4.7 Test Methods.

4.7.1 *Inspection.* Each gage shall be examined externally to determine conformance to the applicable drawings and to all the requirements of this specification not covered by tests.

4.7.2 *Scale error at room temperature.* The gage sensor shall be subjected successively to the pressures given in Table I for the corresponding gage ranges and the readings shall be checked against a standard mercury manometer. The gage scale error shall not exceed the values listed in Table I. Scale error tests shall be conducted in such a way that all pointers move simultaneously with changes in applied pressure. At the completion of each scale error test, pressure shall be applied to each sensor of the gage independently. The corresponding pointer shall move smoothly throughout the scale range without any interference from the other pointer. When the pressure is varied uniformly, the pointer movement shall be uniform and free from backlash.

4.7.3 *Friction error.* The instruments shall be tested for friction at least at every other test point, as applicable for the 75-inch and the 100-inch gages listed in Table I. Pressure shall be applied to bring the pointer approximately to the desired reading and shall then be held constant while two readings are taken. The first reading shall be taken before tapping the instrument, and the second reading after tapping the instrument. The difference between any two readings is the friction error and shall not exceed 0.5 inch of mercury for the 75-inch gage and 0.6 inch for the 100-inch gage. The pointers shall move uniformly without vibrating or tapping the instrument. This test may be accomplished in connection with the test for Scale error at room temperature.

4.7.4 *Position error.* With atmospheric pressure applied, the reading of the gage shall be observed in the normal operating position (plane of dial vertical). The gage shall then be rotated 90 degrees clockwise and 90 degrees counterclockwise from the initial position maintaining the plane of the dial vertical. The gage shall then be tipped forward and backward 45 degrees from the initial position. The readings shall be noted in these positions while the gage is being lightly tapped. No reading shall differ from the reading taken in normal operating posi-

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tion by more than 0.2 or 0.3 inch of mercury for the 75- or 100-inch gage respectively.

4.7.5 *Damping.* A pressure sufficient to produce a scale deflection of the pointer to the 50-inch mark for the 75-inch gage or the 70-inch mark for the 100-inch gage shall be applied to the pressure sensor. The connection shall then be broken suddenly and the time required by the pointer to return to 32 or 35 inches, respectively, shall be recorded and shall not exceed 2 ± 0.5 seconds.

4.7.6 *Seasoning.* The gage shall be subjected to 3000 applications of pressure to produce an indication of from 30 to 65 inches of mercury for the 75-inch gage or 30 to 80 inches for the 100-inch gage at a rate of 10 ± 2 cps. One hour after completion of the Seasoning test the gage shall be subjected to and meet the requirements of the Scale error at room temperature and Friction tests.

4.7.7 *Scale error at high temperature.* The gage shall be tested in accordance with Procedure II of Specification MIL–E–5272 except that the time at high temperature shall be 4 hours. At the end of this period and while still at +71° C., the gage shall be tested as described in the Scale error at room temperature test. Errors shall not exceed those shown in Table I by more than 0.3 inch, 0.4 inch for the 75- or 100-inch gage respectively. Not sooner than 3 hours after return to room temperature the gage shall be subjected to and meet the requirements of the Scale error at room temperature test.

4.7.8 *Scale error—high altitude, low temperature.* The gage shall be subjected to Condition C, Procedure VI of Specification MIL–E–5272. At the end of a 4 hour period and while still at −54° C., the gage shall be subjected to the Scale error and Friction error tests. The scale errors shall not exceed those specified in Table I by more than 0.3 inch or 0.4 inch of mercury for the 75- or 100-inch gage respectively. The friction error shall not exceed 0.7 or 0.8 inch of mercury, respectively. Not sooner than 3 hours after return to room temperature and atmospheric pressure the gage shall meet the requirements of the Scale error at room temperature and Friction error tests.

4.7.9 *Overpressure.* A pressure of 5 inches of mercury in excess of the full scale value shall be applied to the gage for a period of 10 minutes. The pressure shall then be reduced to approximately 2 inches of mercury absolute and maintained for a period of 10 minutes. This pressure shall then be released and the gage shall immediately be subjected to and meet the requirements of the Scale error at room temperature test.

4.7.10 *Vibration error.* The gage shall be vibrated in accordance with Procedure IV of Specification MIL–E–5272. The test shall be conducted with atmospheric pressure applied to the gage. The natural frequency of the gage shall not be within the range of the test frequencies. Pointer oscillation and variation shall not exceed ± 0.6 inch or ± 0.8 inch of mercury for the 75- or 100-inch gage, respectively.

4.7.11 *Vibration failure.* The gage shall be vibrated in accordance with Procedure V of Specification MIL–E–5272. The test shall be conducted with atmospheric pressure applied to the gage. Following vibration the gage shall be subjected to and meet the requirements of the Scale error at room temperature and the Friction error tests.

4.7.12 *Magnetic effect.* The gage shall be revolved about a short bar-magnet compass with the nearest part of the gage 5-½ inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The gage shall be revolved in a horizontal plane which is perpendicular to the axis of the bar-magnet. The gage shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees. At each of these positions the gage shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed one degree.

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4.7.13 *High temperature exposure.* The gage shall be tested in accordance with Procedure II of Specification MIL–E–5272. The gage shall be tested at +71° C. as described in the Scale error at room temperature, test. The scale errors shall not exceed the scale errors shown in Table I by more than 0.3 inch or 0.4 inch of mercury for the 75- or 100-inch gage respectively. Not sooner than 3 hours after return to room temperature the gage shall be subjected to and meet the requirements of the Scale error at room temperature test.

TABLE I. *Scale Errors*

Pressure inch/Hg		Scale error inch Hg, Max	
		75-inch range	100-inch range
30		0.3	0.4
25		0.4	0.5
20		0.4	0.6
15		0.5	0.7
10		0.6	
20		0.4	0.6
30		0.3	0.4
35		0.3	0.4
40		0.3	0.4
45		0.3	0.4
50		0.4	0.5
55		0.4	0.5
60		0.4	0.6
65		0.5	0.6
70		0.6	0.7
75		0.6	0.7
50	75-inch	0.4	
30	gage only	0.3	
80			0.8
85			0.9
90			0.9
95	100-inch		1.0
100	gage only		1.0
50			0.5
30			0.4

4.7.14 *Low temperature exposure.* The gage shall be tested in accordance with Procedure II of Specification MIL–E–5272. At the end of the 72 hour period at −65° C., the gage shall be operated over the entire scale range to determine that the pointer moves smoothly and that no component parts have been damaged. At the end of the final 24 hour period at −54° C., the gage shall be tested as described in the Scale error at room temperature and Friction error tests. The scale errors shall not exceed those specified in Table I by more than 0.3 inch or 0.4 inch of mercury for the 75- or 100-inch gage, respectively. The friction error shall not exceed 0.7 or 0.8 inch of mercury respectively. Not sooner than 3 hours after return to room temperature the gage shall be subjected to and meet the requirements of the Scale error at room temperature test.

4.7.15 *Humidity.* The gage shall be tested in accordance with Procedure I of Specification MIL–E–5272 except that there shall be 5 cycles (120 hrs). Within one hour of removal from humidity the gage shall be subjected to and meet the requirements of the Scale error at room temperature and Friction error tests. There shall be no corrosion of any of the components.

4.7.16 *Fungus resistance.* The gage shall be tested in accordance with Procedure I of Specification MIL–E–5272 except that the test period shall be 14 days. After the testing procedure has been completed the gage shall be examined. Any fungus growth shall be cause for rejection. At the option of the qualifying agency, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled instrument.

4.7.17 *Salt Spray.* The gage shall be subjected to the Salt spray test in accordance with Procedure I of Specification MIL–E–5272. Within one hour of removal from the salt spray the gage shall be subjected to and meet the requirements of the Scale error at room temperature and friction error tests. Twenty-four to forty-eight hours later the gage shall meet the requirements of the Scale error at room temperature and Friction error tests.

4.7.18 *Life Test.* The gage shall be subjected to the same conditions as required by the Seasoning test except that there shall be 100,000 applications of pressure. Not sooner than one hour after completion of the Life test, the gage shall be subjected to the Scale error at room temperature and

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Friction error tests. The tolerance shall be double that allowed in Table I.

4.7.19 *Mounting nuts.* Each attached mounting nut, if furnished, shall have a load of 25 pounds applied in such manner as to tend to pull the nut out of the lug toward the rear of the case. Each insert nut shall be subjected to a torque of 10 pound-inches about the thread axis. The nut shall not be loosened, nor shall it rotate more than a fraction of a turn.

4.7.20 *Mounting lugs.* The instrument case or gage shall be mounted face downward on a movable head of a suitable testing machine with the diameter of the case in a horizontal plane to insure that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug or mounting nut and attached to the pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for 1 minute to each lug along the insert nut axis and toward the front of the instrument. The lugs shall withstand the applied load without fracture.

4.7.21 *Internal examination.* Upon completion of all other tests the gage shall be opened and examined. The examination shall determine conformance with this specification with respect to material and workmanship. Any shortcoming in these respects, or the presence of excessive wear or deterioration brought about as a result of the tests, shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packing

5.1.1 *Level A.* Each gage shall be preserved in accordance with Method 1A8 of MIL–P–116 without preservative compound, and shall be over-boxed in an outer container conforming to PPP–B–636 (Style and Class optional) unless otherwise specified by the Procuring Activity.

5.1.2 *Level C.* Each gage shall be preserved in accordance with Method III of MIL–P–116 and packaged in accordance with MIL–P–7936.

5.2 **Packing.** Packing for Levels A, B, C shall be as specified in contract or order (See para. 6.2) and shall be in accordance with MIL–P–7936.

5.3 **Marking for Shipment and Storage.** Interior and exterior container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** The gages covered by this specification are intended for use in aircraft to indicate the pressure existing in the intake manifolds of the engines.

6.2 **Ordering data.** Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether Sampling Plan is to be omitted, or name of the laboratory where samples shall be submitted.

c. Levels of packaging and packing desired.

d. The dash number of the gage being ordered.

6.3 **Qualification.** With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be elibible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification should be addressed to the Commander, Middletown Air Materiel Area, Olmsted Air Force Base, Pa., the

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activity responsible for qualification, with a copy to the Bureau of Naval Weapons, Navy Department, Washington 25, D.C.

6.4 Provisions of this specification are subject to International Standardization agreement (ABC AIR-STD 10/3). When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:

Army—MO
Navy—WP
Air Force—67
International Interest—(see 6.4)

Reviewer Activity

Army—MO
Navy—WP
Air Force—67

Preparing Activity

Air Force—67

Project Nr. 6620-0072

Review/User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DOD Index of Specifications and Standards.

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